# Consumer AI Scoring of IGCSE Summary Writing: Consistency, Agreement and Construct Validity for Multilingual Learners

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## Abstract

Teachers increasingly paste student writing into consumer AI applications such as ChatGPT and Claude and ask for a mark. Under these conditions the user cannot control sampling or reliably detect changes to the underlying model, so for a task, population and mode of use like those studied here, it is not known how consistently the same script is marked, where AI marks diverge from human judgement, or whether the justifications the tools give respect what the task is designed to measure.

This study examines these questions for the selective summary task of Cambridge IGCSE First Language English (0500, Paper 1, Question 2(a)), marked separately for reading (10 marks) and writing (5 marks). Students in four IGCSE classes at a school in Erbil, Kurdistan Region of Iraq (expected n = 65–80), who use Kurdish and/or Arabic at home, will each summarise two original passages, yielding 130–160 scripts. Two experienced raters will mark the typed, anonymised scripts independently and blind. Each script will then be scored three times through paid consumer subscriptions to ChatGPT and Claude, in separate sessions within a ten-working-day window.

Analyses combine pass-to-pass consistency, single-run agreement statistics (quadratic weighted kappa, intraclass correlations) and a mixed model of tool, passage and pass effects. The criterion-level difference between AI and human marks—organised under the label scoring correction gap—is examined further through a focused, dual-coded analysis of the AI rationales for 8 scripts selected to cover both passages, both criteria, both directions of disagreement and close-agreement cases. Each rationale claim is checked against the source passage, the student's summary and the rubric, to examine whether the tools award plausible marks for reasons the rubric does not support—for example, crediting fluent prose with reading marks that the content does not justify. The study provides evidence to inform further evaluation of AI as a second marker for this task; it does not establish operational thresholds for teacher use.

# Chapter 1. Introduction

## 1.1 Background

Consumer generative AI has entered teachers' marking. Teachers now use general-purpose chat applications for many professional tasks, including marking. In a spring 2025 survey of 2,232 United States public school teachers, 60% reported using an AI tool for their work during the 2024–25 school year, and 16% reported using AI tools to grade or provide feedback (Gallup & Walton Family Foundation, 2025). An analysis of educators' conversations with Claude found that assessing student performance accounted for about 7% of the conversations analysed, that 48.9% of grading-related conversations were automation-heavy, and that surveyed faculty rated grading as the area in which AI was least effective (Anthropic, 2025a). These populations differ from teachers in Iraq or Malaysia, but the figures indicate that using a consumer chat application as a marker is an existing practice.

Consumer access differs from the conditions studied in most scoring research. Research on large language model (LLM) scoring has largely used application programming interfaces (APIs) or controlled settings, which show what models can do with fixed versions and specified sampling parameters. A teacher with a monthly subscription can choose a named model in the model picker but cannot set sampling parameters such as temperature, pin the underlying model version or reliably know when the service behind a model name has been updated. OpenAI's release notes, for example, record successive launches, updates and retirements of the models offered in ChatGPT (OpenAI, n.d.-a), and Chen et al. (2024) found that the behaviour of GPT-3.5 and GPT-4 changed substantially between March and June 2023. Findings established under API conditions therefore do not transfer automatically to consumer use. The present study treats consumer access not as a methodological innovation but as the condition under which teachers actually use these tools, whose scoring properties need their own evidence.

The IGCSE selective summary is a suitable test case. In the 0500 syllabus for examination in 2027–2029, Paper 1 Question 2(a) is a selective summary written in the candidate's own words, as continuous writing of no more than 120 words, carrying 10 reading marks and 5 writing marks (Cambridge International Education, 2024c). The task is a highly constrained, rubric-bound writing task within the qualification. In the study school it is taught in both Grade 9 and Grade 10, and its component skills overlap with those required in English for Academic Purposes (EAP), IELTS, A-Level and IB contexts. Because it rewards both the selection of relevant content points and the quality of the writing, it cannot be marked without the source passage. This makes it a demanding but transparent test case for studying the consistency and human–AI

agreement of AI scoring under consumer-use conditions. To examine whether AI scoring behaviour varies across passages within the same task type, the study uses two passages rather than one.

Second-language learners sitting a first-language examination present a distinctive scoring challenge. The participants use Kurdish and/or Arabic at home and attend a private English-medium school in Erbil. Private English-medium schooling is significant in the region (Hassun, 2022), and second-language learners in such programmes are often assessed against first-language expectations (Carder, 2007). In a first-language examination the language is the construct, so a scorer—human or AI—must distinguish between language that is incorrect and language that is communicatively effective even where it shows features observable in multilingual writers' prose. Messick's (1989) distinction between construct underrepresentation and construct-irrelevant variance frames the concern: an AI scorer that confuses reading and writing constructs—for instance, using fluent prose to justify reading credit that the textual evidence does not support, or overlooking whether the student has selected and transformed the relevant content points—underrepresents the reading construct, while one that penalises language on grounds not supported by the applicable rubric descriptor—rather than on the basis of meaning or accuracy—introduces construct-irrelevant variance. Whether consumer AI tools exhibit either tendency when scoring summaries by multilingual adolescents has not, to the author's knowledge, been addressed in the published literature reviewed in Chapter 2.

The study's setting contributes to geographic diversity in AI scoring research. Virtually all empirical studies of LLM scoring of second-language writing reviewed in Chapter 2 use East Asian populations (Chinese, Japanese, Korean) or, less frequently, Latin American or European learners. Kurdish–Arabic–English multilingual adolescents are absent from this literature. The Kurdistan Region of Iraq, where private English-medium schooling has expanded significantly (Hassun, 2022), represents a growing population of multilingual learners sitting first-language examinations under conditions that have not been studied. The study does not claim generalisability from one school to the region, but it addresses a population underrepresented in the AI scoring literature reviewed in Chapter 2.

Score reliability matters even in classroom assessment. IGCSE summary marks inform teachers' judgements about readiness, setting and predicted grades. If an AI mark varies between submissions, a teacher relying on it may make inconsistent decisions about the same student. Validity concerns "the degree to which evidence and theory support the interpretations of test scores for proposed uses of tests" (American Educational Research Association [AERA] et al., 2014, p. 11), and information on scoring

consistency is essential to estimating score precision (AERA et al., 2014, p. 92). A teacher who accepts an AI mark makes such an interpretation, usually without evidence.

## 1.2 Problem statement

Teachers who use ChatGPT or Claude through consumer subscriptions to mark student writing do so without evidence on three properties of the resulting scores, and without understanding of the nature of AI–human disagreement in this context. For this task, this population and this mode of use, it is not known how consistently the same script is marked when submitted repeatedly in independent sessions (pass-to-pass consistency); whether the tool's scoring properties hold across different passages within the same task type (passage dependence); or what the AI tool's scoring rationale reveals when its mark diverges substantially from human judgement, particularly for second-language learners whose writing may be penalised or credited on construct-irrelevant grounds.

Existing studies of LLM scoring have mostly examined API or research access and essay tasks by university writers (Chapter 2). Repeated scoring, criterion-level comparison and rationale analysis have each been studied (e.g., Liu et al., 2026; Sawaki et al., 2025; Wu et al., 2026), but not, to the author's knowledge, together for an adolescent source-based task with separate reading and writing marks, scored under consumer-subscription access, with AI justifications checked claim by claim against the source passage, the student text and the rubric.

## 1.3 Aim and objectives

The aim is to provide evidence to inform the further evaluation of a discrepancy-review workflow in which a teacher uses a consumer AI tool as a second marker for IGCSE summary writing, and to examine where AI scoring diverges from human judgement and what justifications the tools provide, in a multilingual sample. The objectives are to:

1. estimate the pass-to-pass consistency of ChatGPT and Claude scores for the same summaries scored three times through consumer subscriptions, across two passages;
2. measure agreement between AI and human scores, and locate the correction gap by criterion, score level and passage;
3. analyse the scoring rationales of selected high-divergence and close-agreement cases to examine what justifications the tools provide and how well those justifications accord with the available textual evidence, with attention to source fidelity, construct-irrelevant features, and the treatment of non-standard but meaningful phrasing in this multilingual sample.

The study's synthesis objective is to draw on the findings of objectives 1–3 to assemble descriptive evidence that can inform further evaluation of a discrepancy-review workflow.

## 1.4 Research questions

| RQ | Question | Layer |
|---|---|---|
| RQ1 | To what extent do ChatGPT and Claude, used through consumer subscriptions, give consistent scores to the same summary when it is scored three times in independent sessions, and does consistency differ across passages? | Pass-to-pass consistency |
| RQ2 | How closely do AI scores agree with those of two experienced human raters, and where (by criterion, score level and passage) do they diverge? | Agreement and correction gap |
| RQ3 | What justifications do the AI tools provide for their marks in selected high-divergence and close-agreement cases, how well do those justifications accord with the available textual evidence, and what do they reveal about how the tools handle source fidelity, construct-irrelevant features and non-standard but meaningful phrasing in this multilingual sample? | The scoring correction gap explained |

The discrepancy-review workflow is the study's synthesis objective, not a separate research question. The synthesis draws on the findings of RQ1–RQ3 to assemble descriptive evidence that can inform further evaluation of such a workflow (see Section 3.7).

## 1.5 Significance

**For teachers and schools.** The study will give teachers who already use consumer AI tools descriptive evidence on the consistency and agreement properties of consumer AI scoring, informing their consideration of whether a tool is consistent enough to use as a second marker—under the ecological conditions of ordinary consumer access rather than through an API pipeline that most teachers cannot build. The content analysis of high-divergence cases will illustrate the kinds of scoring justifications AI tools produce and how well those justifications hold up against the textual evidence, and the analysis of how the tools treat non-standard but meaningful phrasing will address a concern relevant to the many international schools where multilingual learners sit first-language examinations.

**For language assessment research.** The study treats the consumer AI application as a rater whose behaviour may vary across passages within the same task type. Repeated scoring across two passages allows within-passage pass-to-pass variation and passage effects to be estimated separately from human–AI disagreement. The scoring correction gap—a proposed analytical label for the criterion-level, rationale-supported description of where AI scores diverge and what justifications the tools offer—describes what a

teacher must understand about an AI mark before acting on it. This label, and the structured evidence-checking procedure that supports it, is the study's descriptive contribution. The study also contributes to geographic diversity: it addresses a population—multilingual Kurdish–Arabic–English adolescents—underrepresented in the literature reviewed.

**For validity theory applied to AI scoring.** The study applies an argument-based approach (Kane, 2013) to a use outside the operational settings for which automated scoring frameworks were developed (Williamson et al., 2012; Xi, 2010), showing which inferences in a teacher's use of an AI mark can be supported in a qualification-relevant classroom context (IGCSE). The reading–writing criterion-separation analysis contributes to the broader question of whether AI scoring introduces construct-irrelevant variance for L2 populations.

## 1.6 Scope and key definitions

The study is bounded to one task (Question 2(a)), two original passages, one school, two consumer AI tools and one scoring wave. It examines use through paid individual subscriptions; free versions of the same applications may offer different models and stricter usage limits, so the findings do not automatically apply to them.

**Consumer AI access (ecological condition).** Use of ChatGPT or Claude through a paid individual subscription to the provider's chat application, without API access. The latest flagship model offered on the paid plan at the start of scoring is selected in the model picker, and the same named model is selected manually for every submission thereafter. The user cannot set sampling parameters, pin the version behind a model name or reliably see model changes; the model name and reasoning mode are therefore logged for every submission. This is a methodological choice motivated by the intended use context: it reproduces the conditions under which teachers typically use these tools.

**Repeatability under consumer-use conditions.** Variation in the scores a single tool assigns to the same script across three independent passes in separate sessions within a ten-working-day window, under the same framing. This estimate includes both within-session random variation and any between-session variation from the service. With three passes, the estimate of within-script variability rests on three scores per script, providing a within-script variance estimate that does not depend on a single difference. Also referred to as pass-to-pass consistency in this proposal.

**Correction gap (scoring).** The AI score minus the mean of the two human raters' scores, so that a positive value indicates AI leniency, analysed by criterion, score level and passage. Scoring correction gap is proposed as an analytical label—informed by prior work on evaluating and revising machine output—for describing, at the criterion

level, the documented difference between what the AI awards and what the human benchmark awards, characterised not only by magnitude but by the reasoning the AI provides. It is not a new theoretical construct; it is a descriptive tool for organising the comparison of AI and human scoring. The prompt framing requests a short reason alongside the mark; these reasons are retained in the response log and are analysed for 8 prespecified scripts (Section 3.7).

**Reading–writing criterion separation.** In a first-language examination, the language is the construct. The central risk is that an AI scorer confuses reading and writing constructs: using fluent, well-formed prose to justify reading credit that the textual evidence does not support, or overlooking whether the student has selected and faithfully transformed the relevant content points (the reading construct), or penalising language features observable in multilingual learners' writing on grounds not supported by the applicable rubric descriptor for the criterion on which the mark is awarded. Because the Cambridge writing descriptors explicitly assess clarity, fluency, organisation, concision and vocabulary, rewarding those qualities under the writing mark is legitimate; construct confusion arises only when a quality is used to justify a mark under the wrong criterion—for example, crediting fluency under the reading mark without supporting content-point evidence. The content analysis examines whether AI scoring rationales show this tendency, judged against the applicable level descriptor for the criterion in question.

**Non-standard but meaningful phrasing.** Wording in a student's summary that departs from conventional English usage—for example, unconventional collocations, article or preposition choices, or verb forms—while still conveying the source meaning accurately. Such wording is identified from the text itself, at claim level, during the evidence-checking procedure (Section 3.7); it is not attributed to any particular home language. Copying from the source, inaccurate content and errors that obscure meaning are coded separately and are not treated as features of multilingual writing. Because the study has no comparison group of monolingual writers, it can describe how the AI tools treat such phrasing within this multilingual sample; it cannot establish that any scoring pattern occurs because students are multilingual, or that it affects them more than other populations.

**Discrepancy-review workflow.** A teacher-facing workflow in which the teacher marks a script first, then submits it to an AI tool, and compares the two marks. The study provides descriptive evidence—consistency statistics, agreement benchmarks, correction-gap profiles and qualitative explanations—to inform further evaluation of this workflow for the IGCSE summary task. It does not test the workflow with teachers or specify when a teacher should trust an AI mark; that requires further research.

# Chapter 2. Literature Review and Conceptual Framework

## 2.1 Summary writing as a reading-into-writing construct

Summary writing integrates comprehension and production. Kintsch and van Dijk (1978) modelled comprehension as the construction of a local text base and a global macrostructure, produced by macro-operators that reduce a text to its gist. Brown and Day (1983) operationalised this reduction as rules for deleting trivial and redundant information, replacing lists with superordinate terms, and selecting or inventing topic sentences; younger students relied on a "copy-delete" strategy, whereas expert summarisers restated the gist in their own words. Hidi and Anderson (1986) showed that summarisation demands depend on the text, on whether the source is available and on whether the summary is written for oneself or for a reader; the IGCSE task is reader-based and source-available.

In L2 summarising, reliance on source wording varies with proficiency, writing experience and task demands. Johns and Mayes (1990) found that lower- and higher-proficiency university ESL students differed significantly in replicating source sentences and in combining idea units across sentences. Shi (2004) found that students writing summaries borrowed more words from source texts than those writing opinion essays. Keck (2006) distinguished degrees of paraphrase, and Keck (2014) found that less-experienced L1 and L2 university writers relied more heavily on source-text excerpts than more-experienced peers. In assessment, Cumming (2013) identified two perils of integrated reading–writing tasks: they confound writing ability with source comprehension, and they make source language hard to distinguish from the writer's own. The 2024 examiner report on the equivalent Cambridge question shows the same weaknesses: less effective summaries copied with minimal rewording, repeated ideas, added comment or exceeded the 120-word guidance (Cambridge International Education, 2024a). These tendencies may occur in the study corpus.

## 2.2 Accessibility, L2 writers and the first-language construct

Messick's (1989) distinction between construct underrepresentation and construct-irrelevant variance frames the central scoring tension for L2 writers in an L1 examination. Where language is not the construct, reducing linguistic difficulty removes irrelevant barrier. But in IGCSE 0500, reading a demanding source and transforming it into one's own words is the construct. A student whose summary preserves meaning through communicatively effective phrasing—even where that phrasing shows features observable in multilingual writers' prose—has demonstrated the reading understanding the task assesses, although the quality of that phrasing may still warrant a lower writing mark; a scorer that withholds reading credit for such

phrasing while crediting a more fluent but less faithful alternative misrepresents the reading construct.

The tension between first-language assessment expectations and multilingual learner populations extends beyond individual classrooms. Gallagher (2008) examined the challenges of assessing English language learners within international school curricula designed for native speakers, arguing that assessment frameworks developed for monolingual populations may systematically disadvantage multilingual students. Sears (2015) found that international school teachers held varied and sometimes contradictory beliefs about how language proficiency should be weighed in content assessment. These findings suggest that the construct validity of first-language English examinations for multilingual learners is an unresolved question in international education. An AI scoring tool that inherits or amplifies the construct-irrelevant features of its training data could deepen this inequity without the corrective judgement that an experienced human marker might apply.

The L2 writing literature shows that source use varies with proficiency. Pecorari (2003) argued that patchwriting in postgraduate writing was typically not intended to deceive and called for a pedagogical response. Hirvela and Du (2013) found that learners could complete paraphrasing exercises with little sense of why paraphrase mattered. Baxter and Neumann (2023) provide evidence on adolescent writers. These findings have a scoring parallel: an AI scorer trained on large corpora of fluent English may implicitly confuse reading and writing constructs, valuing fluent prose over the faithful content-point transformations that the IGCSE rubric rewards for the reading mark. Whether a particular AI judgement constitutes construct confusion depends on the applicable rubric descriptor for the criterion on which the mark is awarded: penalising reliance on source wording under the writing descriptor, which explicitly addresses concise vocabulary use and reliance on the words of the text, is criterion-appropriate; the same judgement applied to the reading mark without content-point evidence is not.

Whether consumer AI tools exhibit this tendency when scoring summaries by multilingual adolescents has not, to the author's knowledge, been addressed in the literature reviewed above. The present study's content analysis of high-divergence cases (RQ3) examines the justifications the tools provide and whether those justifications reveal confusion between reading and writing constructs.

## 2.3 Frameworks for evaluating automated scoring

Validation frameworks for automated scoring are argument-based. Clauser et al. (2002) argued that automated scoring is judged by the same validity criteria as other scoring, but that the parts of the argument needing emphasis depend on the scoring procedure. Xi

(2010) set out validity questions by inference, including whether scoring captures the construct, whether scores resist gaming and whether they generalise and support appropriate decisions. Williamson et al. (2012) proposed evaluating fit to purpose, human–automated agreement, associations with independent measures, generalisability and subgroup consequences, with agreement criteria that include a quadratic weighted kappa (QWK) of at least .70 (see also Doewes et al., 2023). The Standards require empirical support, such as agreement rates with human scorers, before automated scoring is used operationally (AERA et al., 2014, Standard 4.19).

Applying these frameworks to consumer AI scoring requires attention to limited documentation and possible changes in the scoring system. The prompting procedure proposed here does not involve researcher-led model training or fine-tuning: the rubric is supplied in a prompt, which puts scoring within a teacher's reach. However, a consumer chat application is undocumented to its user, may change without notice and may return different scores on resubmission. The present study keeps the frameworks' evidential categories but treats stability over submissions as a property to be established, not assumed.

## 2.4 Empirical evidence on LLM scoring of L2 writing

Agreement with human or benchmark scores is promising but variable. Mizumoto and Eguchi (2023) scored the 12,100 TOEFL11 essays with GPT-3 and reported agreement with human raters sufficient to support human evaluators. Yancey et al. (2023) found that GPT-4 with calibration examples performed almost as well as modern automated writing evaluation in rating short L2 essays, although agreement varied by first language. Tate et al. (2024) found that similarity in average AI and human scores could coexist with appreciable disagreement on individual essays, indicating the need for caution when interpreting aggregate agreement as sufficient for low-stakes use. By contrast, Bui and Barrot (2025) found only weak to moderate correlations between ChatGPT and an experienced human rater on 200 argumentative essays, with low consistency across two scoring rounds.

Repeated scoring does not guarantee repeated scores. Pack et al. (2024) had four LLMs score 119 learner essays on two occasions: GPT-4 showed excellent intrarater reliability, but all models showed performance fluctuations, and agreement with human raters fell slightly over time for GPT-3.5 and GPT-4. Liu et al. (2026) compared five LLMs on 60 essays by non-native university writers and found that some models were highly reproducible across repeated runs and closely aligned with expert raters, while others were inconsistent, compressed scores or underestimated essays. Outside education, setting temperature to zero reduced but did not eliminate non-determinism in ChatGPT code generation (Ouyang et al., 2025), and LLMs configured to be

deterministic still varied in accuracy by up to 15% across runs (Atil et al., 2025). Consumer users cannot set temperature at all.

Scores are sensitive to the prompt and to model change. Several LLMs are highly sensitive to minor prompt formatting (Sclar et al., 2024), and in scoring studies the information supplied in the prompt has affected agreement with human raters (Jin et al., 2025; Sawaki et al., 2025; Yancey et al., 2023). Chen et al. (2024) documented substantial behavioural change in the same LLM service within three months.

## 2.5 Scoring biases and the case for understanding divergence

Automated scorers can weight features differently from human raters. Liu et al. (2026) found that most models' scores were driven by lexical precision and fluency, while others emphasised syntactic complexity. Wu et al. (2026), comparing a GPT-based scorer with 20 human raters on 181 university L2 texts—including summary and translation tasks, assessed without the source texts—across content, organisation, grammar and vocabulary, found the GPT scorer more stable overall but stricter on content and compressed at the upper end of the scale. Such criterion-level differences can be concealed by aggregate agreement. Automated scorers can also reward construct-irrelevant features. Powers et al. (2002) examined circumstances in which e-rater might produce inaccurate evaluations if used as the sole scorer, and Baldwin et al. (2025) showed that strategies a human would recognise as incorrect could inflate LLM scores. LLMs used as judges also show position, verbosity and self-enhancement biases (Zheng et al., 2023). People working with automated aids can fail to detect errors (Parasuraman & Manzey, 2010), and Menze et al. (2026) found more item-writing flaws in items produced through teacher–AI collaboration than in teacher-only items, interpreting this as automation bias.

These findings motivate the study's dual approach: aggregate agreement statistics show how much AI and human scores diverge, while the content analysis of rationales examines what justifications the tools offer and how well those justifications hold up against the textual evidence. A teacher who knows only the magnitude of the gap cannot judge whether to trust or override the AI mark; a teacher who also knows the pattern of divergence—for instance, that the tool's rationales systematically credit fluent but unfaithful summaries—can make a more informed decision.

## 2.6 Rater-mediated assessment

Human raters differ systematically, and this can be modelled. Different raters often assign different scores to the same performance (Eckes, 2015; McNamara, 1996). Wind and Peterson (2018), reviewing 259 studies, found a continuing reliance on aggregate inter-rater reliability and argued for diagnostic, rater-specific indicators. Stability of

rater characteristics across occasions is an established question for human raters (Lumley & McNamara, 1995). Treating each AI tool as a rater allows its severity and agreement to be compared with human raters using the same statistics.

## 2.7 Close precursors and the present study's contribution

**Jiao et al. (2025).** Jiao et al. compared ten LLMs, including ChatGPT 4o and Claude 3.5 Sonnet, with two certified human raters on 120 AP Chinese Language and Culture essays using QWK, Cronbach's alpha and many-facet Rasch measurement (MFRM). The study's limitations include a small sample, a single content domain with generalisation to English writing left for future exploration, only two task types, and a single prompting method. Three further features bear on the present study: the access mode and sampling settings are not reported; no essay is scored repeatedly by the same model; and no model is rescored over time.

**Jin et al. (2025).** Jin et al. used MFRM to compare four human raters with ChatGPT and Claude, each under zero-shot and few-shot prompting, on English essays by 117 Chinese EFL students. The LLM raters were more severe than the humans, few-shot prompting brought them closer to human severity, and the LLM raters were more internally consistent than the humans.

**Sawaki et al. (2025).** Sawaki et al. compared instructor and LLM ratings of 97 summaries by Japanese undergraduates on checklist items about topic sentences, under six prompt designs, with five repeated runs per design. Agreement was satisfactory for low-stakes use for some item–prompt combinations, the LLM tended to be harsher where the two disagreed, and the amount of information in the prompt affected agreement more than the order in which rating and rationale were generated. Sawaki et al. also examined the LLM's rationales to investigate sources of disagreement, identifying problems including inappropriate criterion application and difficulty recognising paraphrase—an analytical approach that the present study extends. This is the closest published evidence on LLM rating of L2 summary content, including repeated scoring.

**Liu et al. (2026) and Wu et al. (2026).** Two recent studies narrow the gap further. Liu et al. examined reproducibility across repeated runs, alignment with expert raters and the linguistic features driving each model's scores for university L2 essays. Wu et al. compared a GPT-based scorer with native and non-native human raters on four analytic criteria, assessing texts as standalone writing without their sources, and found criterion-specific differences in severity. Together with Sawaki et al., these studies mean that repeated scoring and criterion-level comparison are not in themselves new; the present study's distinctiveness lies in how they are combined and applied.

Table 2.1 summarises what each close predecessor establishes and what remains open for the present study.

| Study | Writers and task | What it establishes | What remains open |
|---|---|---|---|
| Jiao et al. (2025) | AP Chinese essays (n = 120); ten LLMs | LLM rater effects compared with certified human raters using MFRM | Access mode and sampling not reported; no repeated scoring; not English writing |
| Jin et al. (2025) | Essays by Chinese EFL students (n = 117); ChatGPT and Claude | LLMs more severe but more internally consistent than humans; few-shot prompting brings them closer to human severity | Essay task, not source-based; no separate reading criterion |
| Sawaki et al. (2025) | Summaries by Japanese undergraduates (n = 97); checklist items on topic sentences | Repeated runs; prompt information affects agreement; rationales reveal criterion misapplication and paraphrase difficulty | Checklist items rather than separate reading and writing marks; university writers |
| Liu et al. (2026) | Essays by non-native university writers (n = 60); five LLMs | Model-specific reproducibility, alignment with experts, and features driving scores | Essay task, not source-based; university writers; no separate reading criterion |
| Wu et al. (2026) | University L2 texts (n = 181), including summary and translation tasks; GPT-based scorer vs 20 human raters | Criterion-specific severity differences; upper-end score compression | Source texts withheld from raters; no separate reading criterion assessed |
| Present study | IGCSE summaries by multilingual adolescents (130–160 scripts); ChatGPT and Claude | To be established | Separate reading and writing marks on a source-based task; three passes under consumer-subscription access; two passages; AI claims checked against source, student text and rubric by two independent coders |

**The present study's contribution.** The study extends rather than replaces prior work. To the author's knowledge, no published study combines the following elements:

1. Separate reading and writing marks on an adolescent, source-based task, so that the study can examine whether an AI tool awards plausible marks while misapplying the construct each criterion is intended to measure—for instance, crediting fluent but unfaithful summaries with reading marks.
2. Consumer-subscription access with repeated scoring across two passages, providing evidence on the mode of use teachers actually adopt.

3. A structured evidence-checking procedure that tests each AI rationale claim against the source passage, the student text and the applicable rubric descriptor, extending Sawaki et al.'s (2025) analytical approach and documented so that other researchers can reuse it.
4. Independent dual coding of AI rationales on a shared, pre-segmented claim list, with inter-coder agreement reported.

The specific models will age. The lasting value lies in showing when score agreement provides misleading reassurance and how criterion-level problems can be identified. The scoring correction gap is an organising label for this analysis; the numerical AI–human difference is an established comparison, and any originality lies in the criterion-level, evidence-checked analysis attached to it.

## 2.8 Conceptual framework

The study adopts an argument-based view of validity (AERA et al., 2014; Kane, 2013; Messick, 1989), in which a teacher's use of an AI mark rests on a chain of inferences. Each research question supplies evidence for one or more inferences, and the discrepancy-review evidence base is the synthesis that follows.

| **Inference (Kane, 2013; Xi, 2010)** | **Claim to be supported** | **Evidence (RQ)** | **Failure that would undermine the claim** |
|---|---|---|---|
| Scoring: stability | The AI mark does not depend on the occasion of submission | Pass-to-pass consistency (RQ1) | Single runs disagree beyond a tolerable range |
| Scoring: agreement with human ratings | The mark agrees with trained human judgement about as well as humans agree with each other | Agreement and correction gap (RQ2) | Systematic leniency or severity by criterion, score level or passage |
| Scoring: passage comparability | The tool's scoring properties are comparable across passages within the same task type | Passage effects (RQ1, RQ2) | Consistency or agreement differs substantially between passages |
| Scoring: construct representation | The AI mark reflects the construct the rubric targets, not construct-irrelevant features | Scoring correction gap examined (RQ3) | AI rationales reveal systematic misapplication of the rubric, confusion of reading and writing constructs, or penalisation of communicatively effective language features |
| Decision | The teacher may retain their mark, or review a flagged discrepancy, based on the evidence | Synthesis (RQ1–RQ3) | Not tested in this study; the synthesis informs further evaluation of this inference |

The addition of construct representation as a separate inference, evidenced through the scoring correction gap analysis (RQ3), distinguishes this framework from a purely statistical agreement study. Agreement can be high while the basis for agreement differs between human and AI raters; where it does, the AI mark may fail on different scripts in unpredictable ways. The reading–writing criterion-separation analysis examines this directly for multilingual learners, focusing on whether the AI tools confuse reading and writing constructs or penalise communicatively effective language features.

# Chapter 3. Methodology

## 3.1 Research design

The study uses a quantitative design with a focused qualitative component. Each student writes one script for each passage (A, B), so scripts are nested within students and passage is a within-student factor; each script answers one passage only. Every script is scored by both tools (ChatGPT, Claude) in three passes under the rubric-embedded framing, so tool and pass are crossed within each script. Three passes, rather than two, provide a within-script variance estimate based on three data points per script rather than a single difference, strengthening the precision of the pass-to-pass consistency analysis (RQ1). Human ratings are the reference for comparison, but neither human rater is treated as ground truth. The analysis plan will be preregistered, for example on the Open Science Framework (OSF), before main-study data collection begins (pilot data inform the plan), so that confirmatory and exploratory analyses are distinguished (Nosek et al., 2018).

## 3.2 Setting, participants and corpus

**Setting and participants.** The study will take place at the candidate's school in Erbil, Kurdistan Region of Iraq, where he has taught IGCSE since 2019. Participants will be students in the four IGCSE classes he teaches, all entered for 0500 First Language English; the expected number is 65–80, known exactly only in the collection year (2027–28).

**Corpus.** Each student will write two handwritten summaries—one per passage—in separate class sessions on different days. Within each class, students will be randomly assigned to write Passage A first and Passage B second, or Passage B first and Passage A second (counterbalanced order). Scripts will be collected immediately after writing. An external anonymiser will replace names with codes and type each script verbatim, so that the candidate never sees the handwriting. The anonymiser will also remove any identifying references students may include in their writing (e.g., personal names, school name, teacher references, identifiable personal anecdotes), replacing them with neutral placeholder text (e.g., [name removed]), reducing the likelihood that the candidate or the second rater can identify individual students from writing content. This supplements the code-based anonymisation and strengthens blinding given the candidate's dual role as teacher and Rater 1. Transcription follows these rules: crossed-out text is retained and marked [crossed out]; illegible text is marked [illegible]; spelling and punctuation are reproduced exactly as written. For marking purposes: text marked [crossed out] is excluded from the word count and is not credited. Text marked [illegible] is likewise excluded from the word count; human raters note [illegible]

passages when they affect creditability of content. These rules will be stated in the human rating guide and in the AI prompt framing. A week later, the anonymiser will re-check all typed scripts against the originals before the corpus is frozen; the candidate never performs this check. All human marking will be done on the typed copies. The typed scripts then form a frozen corpus.

## 3.3 Instruments

**Source passages.** Two original, non-fiction passages (Passage A and Passage B), each of about 300–400 words, written by the candidate and modelled on the 0500 Paper 1 text type, will each contain a clear, countable set of content points and some irrelevant detail, so that selection is actually tested. Original passages have the fewest limitations of the available options: there is no copyright question, as unpublished original texts they are unlikely to appear in any model's training data at the time of first submission, no published mark scheme exists for a model to have memorised, and the content points are fully controlled. The two passages will differ in topic and text structure so that passage effects, if present, are interpretable.

**Structured passage-authenticity check.** Because both study passages are researcher-written, a structured authenticity check will be conducted. Two or three experienced IGCSE colleagues will independently rate both study passages and two or three real published 0500 passages (from pre-2027 examination years) on a short structured rubric covering difficulty, text type, content-point density, vocabulary level and length. The colleagues' ratings will be compared to provide evidence that the study passages are comparable with authentic examination passages. Each passage will also be compared with the specimen Text B for the 2027–2029 syllabus and with the 0500 reading level. The real exam passages are used only in this human comparison exercise and are never submitted to any AI tool.

**Task wording.** The task for each passage will follow the Question 2(a) format: a stated focus, continuous writing in the candidate's own words and no more than 120 words (Cambridge International Education, 2024c), confirmed against the 2027–2029 specimen materials.

**Study rubric and reading keys.** The candidate will write a study rubric that mirrors Question 2(a) and contains no verbatim Cambridge text. It is modelled on the Question 2(a) marking tables in the 2027 specimen mark scheme (Cambridge International Education, 2024b). The rubric has two separate components:

*Holistic reading mark (out of 10).* Reading is marked holistically through five level-descriptor bands, parallel to Cambridge's Table A reading bands for Q2(a). The bands assess the range of relevant ideas the candidate demonstrates understanding of, focus on

the text and task, and selection—from indiscriminate selection or excess material at lower levels to skilful selection that demonstrates an overview at the top. Each band covers two marks (Level 5: 9–10 down to Level 1: 1–2), with 0 for no creditable content. This holistic reading mark is the score analysed in all RQs.

*Reading key.* A separate reading key will be prepared for each passage. Each reading key lists the acceptable content points for that passage and will be included in the rubric-embedded prompt as a guide to what the passage offers. It informs holistic judgement but is not converted into a point-count score.

*Writing mark (out of 5).* The writing mark is awarded through three level-descriptor bands parallel to Cambridge's Table B (Level 3: 4–5; Level 2: 2–3; Level 1: 1), with 0 for no creditable content. The bands address clarity and fluency of expression, effectiveness of organisation, and concise use of a range of appropriate vocabulary, with reliance on the words of the text and lifted sections characterising the lower levels. The same writing rubric applies to both passages.

Each reading key will be drafted by the candidate and checked by the colleague. The full study rubric, including level descriptors and both reading keys, will be included as an appendix in the submitted proposal once the passages have been written.

**Rubric-to-criterion mapping.** Because the study evaluates AI scoring against an IGCSE-aligned study rubric rather than Cambridge's own instrument, the rubric must be shown to preserve the Cambridge criteria. The candidate will prepare a criterion-mapping table (draft in Appendix C) linking each study descriptor to the Cambridge reading or writing criterion it preserves. Two of the IGCSE colleagues involved in the passage-authenticity check will independently judge each mapping as full, partial or absent, with a brief comment. Any descriptor judged partial or absent by either colleague will be revised and re-reviewed before the pilot. This step matters for RQ3: the study can classify an AI judgement as construct confusion only if the study rubric places each quality under the same criterion as Cambridge does.

## 3.4 Human rating

Two human raters will mark every script on both passages: Rater 1 is the candidate, an IGCSE teacher since 2019 and an NCUK EAP examiner and moderator; Rater 2 is an experienced IGCSE colleague.

**Calibration.** Before formal marking begins, both raters will independently mark a small practice set of 5–8 scripts (drawn from previous years or from publicly available student summary writing, not from the study cohort) using the study rubric. They will then meet to compare their marks, discuss disagreements and reach a shared understanding of the rubric's application. This calibration exercise is not a reliability

measure; it is a preparatory step to align rubric interpretation before formal marking begins. If the live pilot (Section 3.6) reveals H1–H2 agreement below ICC(A,1) = .60 on the reading mark, a second calibration round will be conducted using additional practice scripts and focused rubric discussion before main-study marking. This aims to bring the human benchmark to the study's minimum reliability threshold before the main corpus is marked.

**Formal marking.** Both raters will use the study rubric and the appropriate reading key for each passage, marking independently, blind to each other and to all AI scores, in random order, after all scripts have been typed and coded. To further mitigate the candidate's familiarity with student writing, formal marking by both raters will begin no earlier than four weeks after the last writing session, increasing the temporal distance between classroom interaction and blind assessment of anonymised, typed scripts. After independent formal marking is complete, raters will discuss scripts on which their total marks diverge by three or more marks; their original independent marks are retained for all analyses regardless of the discussion outcome. All human marks will be locked before any AI scoring.

If the H1–H2 ICC(A,1) on the reading mark, writing mark or total score falls below .60 for a given criterion on a given passage, this will be reported as a study limitation and the descriptive evidence base will note that the human benchmark for that criterion on that passage did not meet the study's minimum threshold. AI scoring and the prespecified RQ1–RQ2 analyses will still be reported for all criteria. The .60 threshold is a study-specific design choice: Koo and Li (2016) place moderate reliability from .50; .60 is a more conservative floor. If the total-score human agreement falls below .60 but reading and writing each meet the threshold, the component-level agreement findings remain reportable; only the total-score agreement finding is qualified as resting on a weak human benchmark.

Human marks produced. Two raters × (65–80 students) × 2 passages = 260–320 script assessments, each yielding a reading and writing mark.

## 3.5 AI scoring protocol

**Tools and models.** ChatGPT and Claude will be accessed through paid consumer subscriptions, without API access. For each tool, the latest flagship model offered on the paid plan at the start of scoring will be selected in the model picker, and that same named model will be selected manually for every submission, even if the picker's default changes during the scoring window. The model name and reasoning mode will be logged for every submission.

**Framing.** Every submission contains the passage, the task and the student's summary, mirroring the human condition. The rubric-embedded framing contains the mark allocation (10 for reading, 5 for writing), a fixed output line ("Reading _/10; Writing _/5"), a request for a short reason for the marks with reference to the level descriptors, the study rubric and the appropriate reading key for the passage. Rules for refusals and out-of-range or unparseable answers will be fixed in advance. The wording will be fixed before data collection and never changed; a draft is in Appendix A.

**Design summary.** The AI scoring conditions are:

Passage A: ChatGPT × 3 passes; Claude × 3 passes

Passage B: ChatGPT × 3 passes; Claude × 3 passes

**Valid-response rate and retry rule.** The valid-response rate will be reported as an outcome for each tool and passage. Invalid responses—refusals, malformed answers, out-of-range scores—will be characterised by type and by script. All first-attempt responses are retained in full. Where a first attempt is invalid, a single pre-specified retry is permitted and recorded, but retry scores enter only a separate sensitivity analysis. Each comparison uses its own common script set.

**Session controls.** Memory, personalisation and custom instructions will be switched off, as will the model-training setting; temporary chats in ChatGPT, with the unpersonalised option, and incognito chats in Claude will be used (Anthropic, n.d.-a; OpenAI, n.d.-b); and each script will be scored in a fresh chat. Every submission will be logged (Appendix B).

**Passes and timing.** Both tools will be used during the same collection days: within each scoring day, scripts will be scored by both ChatGPT and Claude, so that the two tools are exposed to the same temporal window and any model changes during that window are recorded for both. The three passes will run in separate sessions within a ten-working-day window (approximately four to five hours of submissions per day), with no more than two passes scored on the same day for the same tool. A short window limits exposure to service changes while keeping the daily workload practical. Scoring is scheduled for the school winter break (December 2028 to January 2029), when about two weeks are free of teaching; human marks are locked well before this. The pilot (Section 3.6) will be used to judge whether completing all three passes within the winter break is realistic. All three main passes must be completed within a single bounded window, so that short-term repeatability is not mixed with changes in the AI service over several months. If winter scoring is substantially interrupted, the winter submissions will be treated as preliminary and excluded from the main analysis, and all three main passes will be conducted afresh within the following Newroz break (March

2029) as an alternative window; the switch will be reported. Within each pass, the order of scripts and passages will be randomised.

**Preregistered model-change decision rule.** Selecting the same named model throughout is the primary safeguard. The following rule is a backup for the case in which that model is retired or becomes unavailable during scoring. The model name and any reasoning-mode label displayed in the chat interface will be recorded at every submission. If the recorded model label changes between any two passes for either tool, the following preregistered decision rule applies. The affected tool's data are analysed separately by model version as a primary analysis: pass-to-pass consistency (RQ1) is computed only for scripts that have at least two valid scores under the same recorded model label, and agreement with human raters (RQ2) is reported separately for each model version. If fewer than two passes share the same model label for a given tool, within-version repeatability for that tool is reported as not estimable from the available data. A combined-pass analysis pooling across model versions is reported as supplementary; if the model change is not confounded with the pass boundary, it enters as a covariate in the supplementary model; otherwise, the combined analysis is reported without the covariate and the confounding is noted as a limitation. If the recorded model label changes within a single pass, the change is logged at the submission level; affected submissions are flagged and their inclusion or exclusion is decided at preregistration. This rule addresses observed changes in the displayed model identity; an unchanged label does not guarantee that the underlying service is identical across sessions, and the study cannot separately estimate model and pass effects when a label change is confounded with the pass boundary.

**Stability anchors.** Once per scoring day, three anchor scripts (low, mid and high) selected from the pilot will be scored by each tool in the rubric-embedded framing, yielding six additional submissions per scoring day. Anchor scores are monitored descriptively; no formal threshold is applied. No data will ever be excluded on the basis of an anchor check.

**Workload.** At approximately three minutes per submission:

| Calculation | Submissions |
|---|---|
| Primary: 65–80 scripts × 2 passages × 2 tools × 3 passes | 780–960 |
| Anchors: 3 scripts × 2 tools × 10 scoring days | ~60 |
| Total | ~840–1,020 (~42–51 h over 10 working days) |

## 3.6 Pilot study

**Pilot study.** Before the main data collection, a live pilot will be conducted using a third original passage (not used in the main study) with 10–15 students from a colleague's

class (not the candidate's own students). Each script will be scored by both tools under the rubric-embedded framing, in three passes, to test the full end-to-end workflow—writing, anonymisation, typing, human marking and AI scoring—under realistic conditions. The pilot also tests prompt wording, timing, parsing rules and message caps, and the measured time per submission is used to confirm the main scoring schedule. The pilot supplies the three anchor scripts. Ethics approval for the pilot will be sought as part of the main ethics application. The pilot scripts do not enter the main analysis.

**Preliminary repeatability assessment.** Before preregistration, a preliminary repeatability assessment will be computed from the pilot data: the mean and range of within-script absolute differences across three passes, per tool, to provide an early indication of the score variability a teacher would encounter. This assessment characterises observed score variability; it does not estimate confidence-interval width for the main agreement coefficients. A modest assessment of expected confidence-interval width for QWK and ICC at the study's sample size will be discussed with a statistical adviser before preregistration. The preliminary repeatability assessment does not replace the formal RQ1 analysis. Any formatting or parsing change made after the pilot must be checked to ensure it does not alter scoring instructions; the supervisor and colleague will jointly make that determination and record the decision in the preregistration or pilot log.

## 3.7 Analysis

**Analysis hierarchy.** The primary analyses are organised by research question: RQ1—pass-to-pass consistency; RQ2—agreement and the correction gap; RQ3—content analysis of high-divergence rationales. The synthesis draws on all three RQs. Holm (1979) correction is applied within each family of primary mixed-model contrasts. With three passes, the ICC(1,1) per condition rests on three scores per script, providing a within-script variance estimate that does not depend on a single difference. The study reports ICC(1,1) as a tool-level aggregate and supplements it with per-script standard deviations across the three passes.

**Primary and secondary criteria.** The reading mark (/10) is the primary criterion for all analyses in RQ1 and RQ2. Its 11-point scale (0–10) offers more possible score categories than the writing scale, which may—but need not—yield greater observed variance and more precise agreement estimates. The writing mark (/5) is analysed as a secondary criterion: its 6-point scale (0–5) is more likely to compress observed variance and limit the precision of agreement statistics. All analyses are reported for both criteria, but interpretive weight is placed on the reading mark. The total mark (/15) is reported in supplementary tables only. Treating reading as primary is an analytical decision

motivated by the structure of the assessment—the reading mark carries two-thirds of the available marks and has the longer scale—not a requirement of the marking scheme.

**Units, missing data and uncertainty.** Agreement statistics (QWK, intraclass correlation coefficient [ICC], exact and adjacent agreement) are computed separately for reading (/10), writing (/5) and the total (/15), and separately for each passage. Analysis is complete-case by script: a script missing a human mark or a valid AI score in a given condition is excluded from analyses of that condition, and attrition is reported. Agreement statistics will have bootstrapped 95% confidence intervals (2,000 replications). The resampling unit is the student: each resampled student brings both passage scripts and all repeated scores, preserving the within-student dependence. The 780–960 AI scores are repeated measurements of 130–160 scripts from 65–80 students; they are not 780–960 independent observations, and the precision of agreement estimates is governed mainly by the number of students. At this sample size, confidence intervals for QWK and ICC are expected to be moderately wide, so results will be interpreted through their intervals rather than point estimates. The expected interval width will be estimated with the statistical adviser before preregistration.

**Primary result.** Because a teacher receives a single response, the primary agreement result is the agreement of a single AI run with the humans: QWK and ICC are computed for each pass separately and reported as the mean and range across the three passes, for both passages. Agreement based on the mean of the three passes is a sensitivity check only.

**Agreement statistics.** Weighted kappa uses quadratic weights (Cohen, 1968). Human–AI agreement uses ICC(A,1) (two-way mixed effects, raters fixed, absolute agreement, single measures), computed separately for reading, writing and total on each passage. Each tool is compared with each human rater separately (AI–H1, AI–H2), so that the result is directly comparable with the H1–H2 baseline; QWK is computed the same way. Agreement across passes uses a one-way random-effects, single-measures ICC. ICC confidence intervals will be interpreted against the bands proposed by Koo and Li (2016): below .50 poor, .50–.75 moderate, .75–.90 good and above .90 excellent. Bland–Altman plots will show mean differences and limits of agreement (Bland & Altman, 1986). QWK will be interpreted against the human–human baseline and the operational thresholds of Williamson et al. (2012).

**Mixed model.** Using rubric-embedded scores, fitted separately for reading and writing: score ~ tool × passage + pass + (1|student) + (1|student:passage), where pass has three levels (1, 2, 3), using lme4 in R (Bates et al., 2015). Anchor scores are reported descriptively. If a mid-scoring-window model change occurs and is not confounded with the pass boundary, the change enters as a covariate; otherwise it is reported

descriptively and the limitation is noted. Convergence, singular fits and boundary estimates will be reported. Two specification questions will be reviewed with the statistical adviser before preregistration: whether a tool × pass interaction is needed to allow pass behaviour to differ between tools, and whether a linear model is adequate for the bounded 0–10 and 0–5 scales or whether an ordinal (cumulative link) mixed model is preferable. If the model fails to converge, a preregistered fallback will be used: within-script SDs across the three passes will be computed per script and summarised across scripts within each tool/passage cell as a descriptive substitute. This fallback is acknowledged as partial: within-script SDs characterise per-script variability but do not replace the systematic fixed-effect estimates (tool, passage, pass) that the mixed model provides.

**Correction gap.** The correction gap is the AI score minus the mean of the two human raters, computed per pass and per passage and summarised as mean, standard deviation and range across the three passes. The three-pass design allows the correction gap's variability across passes to be estimated, rather than characterised by a single difference. Score level is defined by the mean of the AI and human scores (Bland & Altman, 1986), because grouping by the human score alone produces a regression-to-the-mean artefact.

**Passage effects.** To examine whether AI scoring properties vary across passages, key statistics (ICC(1,1), QWK, mean within-script SD, correction gap) will be computed separately for each passage and compared. The passage factor in the mixed model tests differences in AI scores between the two passages; whether scoring performance differs is established by agreement, correction gaps and within-passage repeated-score differences. Two passages support comparison between these two passages; they do not establish broad generalisability across IGCSE texts. A raw score difference between passages does not by itself indicate a passage-related AI scoring weakness: if students genuinely find one passage harder, both human and AI scores may be lower. Conclusions about passage-related scoring performance will therefore draw on AI–human correction gaps, agreement with human ratings and within-passage repeated-score differences rather than on raw score differences alone.

**Focused content analysis of AI scoring rationales (RQ3).** Eight scripts—four per passage—will be selected for content analysis using a prespecified rule designed to cover both passages, both criteria, both directions of disagreement and cases where marks agree. Discrepancies are computed from the Pass 1 rubric-embedded scores as the signed AI-minus-human-mean gap, expressed as a proportion of the scale (reading gap/10; writing gap/5), for each tool. A script is eligible only if both tools' Pass 1 responses are valid and contain a usable rationale (at least one claim about the

summary); ineligible scripts are skipped and the number skipped is reported. Within each passage, eligible scripts are selected in this order, each from the scripts not yet selected:

1. Reading discrepancy: the script with the largest absolute proportional reading gap for either tool.
2. Writing discrepancy: the script with the largest absolute proportional writing gap for either tool.
3. Opposite direction: the script with the largest absolute proportional gap on either criterion whose sign is opposite to the reference direction (AI under-scoring if the reference is over-scoring, and vice versa). The reference direction is the sign of the gap that led to script 1's selection; if that gap is zero, or if two gaps of equal size and opposite sign tied for it, AI over-scoring is used as the reference. If no script has a gap in the opposite direction, the next-largest gap on either criterion is selected and this is reported.
4. Closest available agreement: the script with the smallest maximum absolute proportional gap across both tools and both criteria. Its actual gaps are reported, since the closest available case is not necessarily close in absolute terms. This case tests whether the AI's justification is sound even when its mark matches the human benchmark—agreement can rest on reasoning the rubric does not support.

Ties are broken alphabetically by script code, and the actual discrepancy values of all selected scripts will be reported. The selection is carried out mechanically on the score spreadsheet before any rationale is read. For each selected script, one rationale per tool is analysed—the Pass 1 rubric-embedded response—yielding 16 rationale texts (8 scripts × 2 tools), provided enough eligible scripts exist. With eight scripts, the findings are illustrative evidence about construct representation in the selected cases; they do not estimate how often such problems occur across the corpus. The rationales will be coded thematically. The coding scheme is informed by three sources: (a) prior research on AI scoring biases (Baldwin et al., 2025; Powers et al., 2002; Zheng et al., 2023), (b) the summary-writing and source-use literature (Brown & Day, 1983; Keck, 2006; Kintsch & van Dijk, 1978), and (c) the reading–writing criterion-separation framework, focusing on whether the tools confuse reading and writing constructs (Messick, 1989). Anticipated codes include:

> *Source fidelity errors:* AI credits a content point the student did not include, or misses one they did include; AI fails to detect meaning-changing omission or distortion

*Construct-irrelevant rewarding:* AI rewards length, vocabulary sophistication or stylistic features not supported by the applicable descriptor for the criterion on which the mark is awarded. (Rewarding fluency under the writing descriptor, where clarity and fluency are assessed, is not itself construct-irrelevant.)

*Treatment of non-standard but meaningful phrasing:* AI penalises non-standard but meaningful phrasing (as defined in Section 1.6) on grounds not supported by the applicable rubric descriptor for the criterion in question, when a human marker credits the same phrasing as effective transformation

*Rubric misapplication and construct confusion:* AI explicitly applies a level descriptor to the wrong criterion—for example, citing fluency or style as the reason for reading credit, or penalising content selection under the writing mark. Construct confusion is coded only when the rationale demonstrably misapplies the relevant descriptor; a human–AI disagreement alone does not establish an AI error

*Own-words transformation:* AI evaluates paraphrase quality differently from human raters; AI does not distinguish synonym substitution from structural reformulation; AI fails to penalise near-copying under the writing descriptor that addresses reliance on source wording. Penalising source copying is judged against the applicable descriptor, not automatically classified as unfair treatment of multilingual writing

**Evidence-checking procedure.** For each of the 8 selected scripts, a structured evidence record will be compiled before coding begins. For each AI rationale claim (e.g., "the student included point X" or "the summary lacks own-words transformation"), the record documents: (a) the AI's specific claim; (b) the relevant student wording and corresponding source-passage evidence; (c) the applicable reading or writing level descriptor; and (d) a judgement using four defined categories: supported (the claim is borne out by the student text and source); contradicted (the claim is demonstrably false—for example, crediting a content point the summary does not contain); insufficient explanation (the claim is too general to show the basis of the mark—for example, "clear and fluent" offered alongside a reading mark); and ambiguous (the evidence allows more than one defensible reading). A rationale that omits content evidence is not, on that ground alone, treated as evidence of mis-scoring: a short explanation may accompany a defensible mark. Contradicted claims are coded under the relevant evidence-error category; for example, crediting a content point the student omitted is a source-fidelity error. Construct confusion is reserved for demonstrable misuse of one criterion to justify the other criterion's mark—for example, citing fluency as the reason for reading credit. The second human rater will independently examine the 8 selected

scripts and their AI rationales, completing the same evidence record, before seeing the candidate's judgements. The raters' original independent marks are preserved and not revisited. This procedure grounds the content analysis in verifiable textual evidence rather than impressionistic comparison.

**Important caveat.** AI-generated rationales are reported justifications for the marks, not verified accounts of the computational process that produced the scores. The content analysis treats them as the justifications a teacher would read and act on, not as explanations of the model's internal scoring mechanism.

Coding will be conducted independently by both the candidate and the second human rater using the same coding scheme. Coding will be organised in Atlas.ti. Both coders will complete the full evidence-checking procedure for all 8 selected scripts (16 rationale texts) before comparing their coding. The coding unit is the individual claim within a rationale text. Before independent coding begins, the candidate will segment each rationale into its constituent claims and prepare a numbered claim list; the second coder will receive this same pre-segmented list, so that both coders' present/absent decisions refer to identical units. For each claim, each thematic code is treated as a separate present/absent decision: a single claim may receive more than one code (e.g., both construct-irrelevant rewarding and rubric misapplication), and inter-coder agreement is calculated on each per-claim binary decision. Inter-coder agreement will be reported as percentage agreement and Cohen's kappa, calculated before discussion, on these per-claim thematic-code decisions. Disagreements will then be discussed and resolved where possible. The discussion is not recorded; instead, a written log records, for each disputed claim, both coders' initial decisions, the final decision (or "unresolved") and a one-line reason. Unresolved cases are reported. The supervisor will review the final coding of three selected scripts as an additional credibility check. This dual-coding procedure strengthens the qualitative strand's trustworthiness and addresses a limitation common in prior research on AI scoring rationales, where coding was conducted by a single researcher.

Codes will allow for defensible alternative judgements (where either the AI or the human interpretation could reasonably be correct) and unresolved cases (where the textual evidence does not clearly favour either). The analysis is a focused component of the study; it does not change any quantitative result but provides interpretive context for the correction gap (RQ2) and the reading–writing construct-separation question.

**Synthesis.** The synthesis draws on RQ1–RQ3 to: (a) summarise pass-to-pass consistency and agreement with human raters, including the direction and magnitude of the correction gap, by tool, passage and criterion (from RQ1 and RQ2); (b) describe the rubric-related concerns illustrated by the selected cases (from RQ3); and (c) identify the

questions that require further validation before classroom adoption. The quantitative results support comparisons of consistency and agreement across conditions; the selected cases illustrate how problems arise but cannot rank how often they occur in each condition. It does not produce operational thresholds or decision rules; it provides the descriptive evidence a teacher or school would need to make an informed decision about whether and how to use a consumer AI tool as a second marker for this task. Whether teachers find the correction-gap evidence interpretable and actionable is a question for future research.

**If discrepancies are small.** If AI–human agreement is high across both tools and both passages, and the content analysis reveals no systematic patterns of construct confusion or construct-irrelevant scoring, this outcome is reported as evidence that consumer AI scoring of this task, under the study's conditions, agreed closely with human judgement. The study's contribution in that case lies in having established the agreement level under ecological conditions, not in having found a problem. If anticipated patterns (e.g., confusion of reading and writing constructs, penalisation of multilingual learners' language features) do not appear in the high-discrepancy rationales, this is reported as a non-finding in the selected cases. Because only eight scripts are examined, such an absence would not demonstrate that construct problems are absent across the corpus. The preregistered analysis plan will include this contingency.

**Time-based workload comparison.** Observed timings for human marking and AI scoring will be reported briefly to inform practical feasibility.

| RQ | Analysis |
|---|---|
| RQ1 | Primary: one-way random-effects ICC(1,1) across three passes per tool, passage and criterion (reading mark primary; writing mark secondary). Supplementary: per-script SD and range across three passes; exact and adjacent agreement; passage comparison |
| RQ2 | Primary: single-run QWK and ICC(A,1), AI–H1 and AI–H2 (mean and range across passes), per passage. Supplementary: human–human baseline; correction gap per pass and passage; Bland–Altman plots |
| RQ3 | Focused content analysis of AI rationales for 8 scripts (4 per passage: reading discrepancy, writing discrepancy, opposite direction, close agreement): claim-level evidence checking and dual-coded thematic coding with inter-coder agreement, with attention to source fidelity, construct-irrelevant features, construct confusion (reading vs. writing) and non-standard but meaningful phrasing |
| Synthesis | Descriptive evidence to inform further evaluation of a discrepancy-review workflow, drawing on RQ1–RQ3, by tool, passage and criterion; time-based workload comparison |

## 3.8 Ethics and data protection

**Approvals and consent.** The study requires USM ethics approval and the principal's permission. No data will be collected before ethics approval is in place. Whether permission from the Kurdistan Regional Government (KRG) Ministry of Education or

the school's owner is also needed will be checked before enrolment. Parental consent and student assent will be obtained independently of the teacher–student relationship; consent forms will make clear that participation has no bearing on grades, and consent collection will not be conducted by the candidate in his teaching role. Students may withdraw at any time up to the corpus-freezing date. Up to that date, the linkage key (the record matching student identity to script code) is held by the anonymiser and never seen by the candidate. After the corpus-freezing date, the linkage key is destroyed. Students will be informed of this deadline in the consent form and again before typing begins.

**Data access and storage.** Original handwritten scripts are held by the anonymiser only. Linkage records are held by the anonymiser only and destroyed on the corpus-freezing date. Typed scoring texts (coded, de-identified transcripts) are accessible to the candidate, the supervisor, and the second human rater. These are stored in encrypted form on USM-approved research storage. AI response logs and screenshots are accessible to the candidate and supervisor only, except that the second human rater will have access to the selected, de-identified AI responses for the 8 scripts used in the evidence-checking procedure (Section 3.7). Disagreements between the two evidence records will be documented in an evidence-review log, preserving both initial independent judgements alongside any agreed or unresolved outcome. All materials are retained for the period specified in the USM research data management policy. The anonymiser remains the continuing custodian of potentially re-identifiable originals; the ethics application will specify how access to these materials is audited.

**Data in AI applications.** Scripts will be coded, de-identified and typed before entering any AI application, and the model-training setting will be off in both. ChatGPT temporary chats are not used for model training and may be kept for up to 30 days for safety purposes (OpenAI, n.d.-b). Claude's incognito chats are not used for training, even when model improvement is enabled (Anthropic, n.d.-a, n.d.-b). These terms will be checked at the time of data collection.

**Insider research.** The candidate occupies three roles: the students' teacher, the principal researcher and Rater 1. This triple role is the study's most significant ethical and methodological tension, and is mitigated at multiple levels. First, the consent process will make clear that study participation has no bearing on grades, and consent collection will not be conducted by the candidate in his teaching role. Second, all scripts are anonymised, typed verbatim and stripped of identifying content by an external anonymiser before the candidate sees any script. Third, scripts are marked in randomised order, blind to student identity and to all AI scores, with marks locked before AI scoring begins. Fourth, formal marking begins no earlier than four weeks

after the last writing session, increasing temporal distance between classroom interaction and assessment. Fifth, the second human rater provides an independent, fully blind benchmark: where the candidate's marks diverge systematically from the second rater's, this is detectable in the H1–H2 agreement statistics. Sixth, the candidate will not conduct the re-checking of typed scripts against originals; this is the anonymiser's responsibility. Despite these mitigations, the candidate's familiarity with his students' writing abilities cannot be fully eliminated. This is acknowledged as a limitation; the second rater's marks serve as the primary check against unconscious bias.

# Chapter 4. Schedule, Outputs and Limitations

## 4.1 Timeline

The candidature runs from September 2027 to August 2031.

| Period | Work |
|---|---|
| Before enrolment (by Aug 2027) | Prepare ethics documents |
| Sep–Dec 2027 | Submit ethics application in the first weeks of candidature (approval required before the January pilot); finalise both study passages, rubrics and reading keys; colleague review; structured passage-authenticity check and criterion-mapping review; obtain KRG permission if needed |
| Jan 2028 | Fix the prompt wording. Run live pilot on third passage with colleague's class (~10–15 students, three passes per tool, ~60–90 AI submissions plus third passage preparation, recruitment, transcription and human marking). Compute preliminary repeatability assessment. Check H1–H2 agreement on pilot data; if reading-mark ICC(A,1) < .60, conduct second calibration round before main-study marking. Discuss expected CI width with statistical adviser. Use results to finalise anchors and timings. Fix and preregister the analysis plan before main-study data collection. |
| Feb–May 2028 | Students write summaries of both passages in separate sessions (Feb–Mar); anonymiser codes, types and re-checks all scripts before corpus freezing; locked human marking of all scripts (Apr–May) |
| Jun–Nov 2028 | Summer supervision visit (Penang); refine literature review; prepare and test analysis scripts on pilot data |
| Dec 2028 – Jan 2029 (winter break) | AI scoring (both tools interleaved on the same days; three passes per tool within ten working days); anchor checks; apply model-change decision rule if applicable. Newroz 2029 break (March) reserved as an alternative window for the entire scoring exercise if winter scoring is substantially interrupted |
| Jan–Aug 2029 | Analysis; dual-coded content analysis of selected rationales; Papers 1–2 drafted |
| Sep 2029 – Aug 2031 | Thesis writing; optional third paper if data warrant |

## 4.2 Planned outputs

Two primary papers are planned; a third is possible if the data support it.

| Paper | Content | Target journals | Status |
|---|---|---|---|
| 1 | Pass-to-pass consistency and passage dependence of consumer AI scoring of IGCSE summaries (RQ1) | Language Testing; Language Assessment Quarterly | Primary |
| 2 | Human vs. AI rater agreement and the scoring correction gap examined (RQ2, RQ3, Synthesis) | Assessing Writing; Computers and Education: Artificial Intelligence | Primary |
| 3 | Reading–writing construct separation in AI scoring of multilingual learners' summaries: a focused analysis of how AI tools handle source fidelity and language features (extending RQ3 beyond the scoring-agreement focus of Paper 2) | Journal of Second Language Writing; TESOL Quarterly | Optional: only if the RQ3 data yield sufficient depth on construct confusion to sustain a standalone paper distinct from Paper 2 |

In addition, the frozen corpus of coded, typed scripts will be documented for replication. The typed corpus is de-identified: identifying content is removed and the linkage key is destroyed at the corpus-freezing date. The original handwritten scripts, which remain potentially re-identifiable, are held separately by the anonymiser (Section 3.8). Rescoring of the typed corpus by future models may therefore be possible, subject to the scope of the original consent, ethics approval and institutional data-management requirements.

## 4.3 Limitations and mitigations

| Limitation | Response |
|---|---|
| Teacher, researcher and rater are the same person | Independent second rater; anonymiser types and de-identifies scripts (names, school references, identifiable anecdotes removed); blind, locked marking of coded, typed scripts in random order; four-week delay between last writing session and formal marking; second rater provides fully independent benchmark; H1–H2 agreement statistics detect systematic divergence. Complete elimination of familiarity bias is impossible; acknowledged as a limitation. |
| One school, one task | A deliberately controlled design; passage effects are estimated with two passages; claims are bounded to this task type and school context |
| Two passages, both researcher-written | Structured authenticity check against real 0500 passages by independent colleagues; passage comparison within the study |
| 130–160 scripts (65–80 students × 2 passages) | Bootstrapped 95% CIs reported; expected precision discussed in preregistration |

| Limitation | Response |
|---|---|
| Three passes limit repeatability precision relative to operational AES systems | Three passes provide a within-script variance estimate based on three data points rather than a single difference; individual-script precision remains limited, and the expected precision of tool-level ICC(1,1) will be assessed with the statistical adviser before preregistration. More passes would improve precision but exceed the study's practical constraints. |
| No comparison group for language background | The study describes how the AI tools treat non-standard but meaningful phrasing within a multilingual sample; it cannot attribute scoring patterns to students' language background or show differential impact relative to other populations |
| Study rubric adapted from, not identical to, the Cambridge mark scheme | Criterion-mapping table independently reviewed by two IGCSE colleagues (Section 3.3; Appendix C); findings are framed as evidence about AI scoring against an IGCSE-aligned rubric |
| Counterbalanced but not fully crossed passage order | Random assignment of A–B / B–A within each class controls for order at the group level; individual order effects cannot be estimated with only two passages |
| Two tools, not the full market | ChatGPT and Claude are two widely used consumer AI chat applications; other tools are excluded; claims are limited to the tools studied |
| Single scoring wave | Temporal drift is not addressed; this is left for future research, which could rescore the same frozen corpus |
| Prompt sensitivity | The study uses a single fixed framing; prompt-sensitivity effects are acknowledged as a limitation and left for future research. |
| Content analysis limited to 8 scripts | Selection rule covers both passages, both criteria, both directions of disagreement and close agreement; two coders work independently from the same pre-segmented claim list, and inter-coder agreement is reported; the supervisor reviews a sample. Findings are reported as illustrative evidence about the selected cases, not as prevalence estimates. |
| Discrepancy-review workflow not tested with teachers | The study provides scoring evidence only; whether teachers find the correction-gap evidence interpretable and actionable is left for future research. |
| Anchor monitoring descriptive only | Anchor scores are monitored descriptively without a formal threshold; they provide a record of day-to-day stability but do not trigger formal action |

## 4.4 Expected contribution

The study is expected to contribute in three ways.

Empirically, it will estimate pass-to-pass consistency, human–AI agreement and the scoring correction gap for two widely used consumer AI tools on an authentic qualification-relevant secondary assessment task (IGCSE) across two passages, under the ecological conditions of consumer access. Its central empirical contribution is evidence on how consistent consumer AI scoring of an IGCSE selective summary is, how closely it aligns with human judgement on each criterion, and, where it diverges,

what the pattern of divergence reveals—evidence that can inform, but does not by itself establish, the use of AI as a second marker.

Conceptually, it will propose the scoring correction gap as an analytical label for describing, at the criterion level, what a teacher must understand about an AI mark before acting on it, supported by a structured evidence-checking procedure that verifies AI justifications against the student text, source passage and rubric descriptors. The reading–writing criterion-separation analysis will contribute to the broader question of whether AI scoring confuses reading and writing constructs or introduces construct-irrelevant variance for multilingual populations on source-based tasks.

Practically, it will provide descriptive evidence—consistency statistics, agreement benchmarks, correction-gap profiles and qualitative explanations—to inform further evaluation of consumer AI tools as second markers for this task. The evidence is intended as a foundation for further validation, including studies with teachers, rather than immediate operational adoption.

# Appendix A. Draft Prompt Framing

**Status: DRAFT.** This wording is illustrative only. The final wording will be fixed before data collection and then never changed. Square-bracketed items are placeholders.

**Rubric-embedded framing**

> Below are a reading passage, a summary task, and a student's summary written in response to the task.
>
> PASSAGE: [PASSAGE]
>
> TASK: [TASK]
>
> STUDENT'S SUMMARY: [SUMMARY]
>
> The student's summary may contain text marked [crossed out] or [illegible]. Text marked [crossed out] is excluded from the word count and is not credited. Text marked [illegible] is likewise excluded from the word count; note any [illegible] passages when they affect the creditability of content.
>
> Mark this summary. It is marked out of 10 for reading and 5 for writing.
>
> Use the following marking rubric: [RUBRIC]
>
> Award the reading mark holistically using the level descriptors in the rubric above. The reading key below lists indicative content points to help you understand what the passage offers; treat it as a guide, not a checklist to convert into marks. Other valid ideas not listed here may be credited. [READING KEY]
>
> Give your marks on the first line in exactly this form: Reading _/10; Writing _/5
>
> Then give a short reason for your marks, with reference to the level descriptors.

*Fixed in advance:* the handling of refusals and of out-of-range or unparseable answers, and the parsing rule for the output line.

# Appendix B. Fields of the Session Log

| Field | Description |
|---|---|
| Submission ID | Unique identifier for each submission |
| Date and time | Date and time of submission (with time zone) |
| App and plan | ChatGPT or Claude; interface (web or mobile); subscription plan |
| Model name | Named model selected in the picker, as displayed (the same model throughout unless unavailable) |
| Reasoning mode | Reasoning or thinking setting as displayed |
| Model change | Whether the displayed model label differs from that at the start of the scoring window (used as a covariate only where separately estimable from pass) |
| Framing | Rubric-embedded |
| Passage | A or B |
| Pass | 1, 2 or 3 |
| Randomised position | Position of the submission in the pass's randomised order |
| Script code | Anonymised script code, or anchor code |
| Anchor flag | Whether the submission is an anchor check |
| Settings check | Memory, personalisation, custom instructions and training off; temporary or incognito chat used (yes/no) |
| Full response | Complete text of the AI response |
| Extracted scores | Reading and writing marks parsed from the output line |
| Response status | Valid, refusal, out-of-range or unparseable, handled by the pre-set rules |
| Screenshot | File reference |
| Anomalies | Message-cap interruptions, regenerated responses or other irregularities |

# Appendix C. Draft Criterion-Mapping Table

**Status: DRAFT.** Cambridge criteria are paraphrased from Table A (Reading) and Table B (Writing) for Question 2(a) in the 2027 specimen mark scheme (Cambridge International Education, 2024b), not quoted. Bracketed codes are the assessment objectives the syllabus states Question 2(a) tests: reading R1 (explicit meanings), R2 (implicit meanings and attitudes) and R5 (selecting and using information for specific purposes); writing W1 (articulating what is thought, felt and imagined), W2 (organising and structuring ideas) and W3 (range of vocabulary and sentence structures appropriate to context) (Cambridge International Education, 2024c). The mapping is designed not to move any quality from one criterion to the other. The final mapping will be completed once the study rubric is written, and judged independently by two IGCSE colleagues as full, partial or absent (Section 3.3).

| Cambridge Q2(a) criterion (paraphrased) | Study rubric element | How the element preserves the criterion | Reviewer judgement |
|---|---|---|---|
| **Reading:** understanding of a range of relevant ideas (R1, R2) | Reading level descriptors; reading key | Bands describe the range of relevant ideas understood and how accurately they represent the source meaning; the reading key lists the content points the passage offers | Full / partial / absent |
| **Reading:** selection of relevant ideas, from indiscriminate selection or excess material to careful and skilful selection (R5) | Reading level descriptors | Bands judge which ideas are selected and whether they are relevant. Borrowing source wording is not treated as poor selection; it is assessed under writing | Full / partial / absent |
| **Reading:** focus on the text and task (R5) | Task wording; reading key; reading level descriptors | The reading key includes only points relevant to the task focus; bands credit focus on that aspect | Full / partial / absent |
| **Reading:** overview demonstrated through selection (R5) | Reading level descriptors | Level 5 requires relevant ideas skilfully selected to demonstrate an overview; Level 4 requires some evidence of an overview. No separate requirement to group or link points is added; organisation is assessed under writing | Full / partial / absent |
| **Reading:** holistic banded judgement out of 10 | Five two-mark levels (1–2 to 9–10) plus 0 | Same band structure and holistic approach; 0 for no creditable content; the reading key guides but is not converted into a point count | Full / partial / absent |
| **Writing:** expression: clear and fluent at the top level (W1, W3) | Writing level descriptors | Bands describe how clearly and fluently the summary reads | Full / partial / absent |

| Cambridge Q2(a) criterion (paraphrased) | Study rubric element | How the element preserves the criterion | Reviewer judgement |
|---|---|---|---|
| **Writing:** organisation: from limited to effective (W2) | Writing level descriptors | Bands describe how effectively the summary is organised | Full / partial / absent |
| **Writing:** reliance on the words of the text; lifted sections at the lowest level (W3) | Writing level descriptors | Bands credit transformation of source language and reduce credit for extended reliance on source wording; necessary terms, names and technical words with no reasonable alternative are not penalised | Full / partial / absent |
| **Writing:** concise use of a range of appropriate vocabulary (W3) | Writing level descriptors | Bands credit concise expression and a range of appropriate vocabulary | Full / partial / absent |
| **Writing:** holistic banded judgement out of 5 | Three levels (4–5, 2–3, 1) plus 0 | Same three-level band structure; 0 for no creditable content | Full / partial / absent |
| **Task:** continuous writing of no more than 120 words | Task wording; marking rules | Same word limit and format; the handling of text beyond the limit follows the study rating guide | Full / partial / absent |

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